

# Transformer: The Next Big Thing

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## Introduction: Transformer

Transformer is a deep learning architecture that replaces Recurrent Neural Networks (RNNs).

LSTM (Long Short-Term Memory) is a type of RNN with 3 layers and 2 hidden states.

Transformer is a deep learning architecture that replaces RNNs.

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2017 Google Research Attention Is All You Need Transformer (Self-Attention)

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## Transformer: Self-Attention

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```
Python: import torch import torch.nn.functional as F def self_attention(Q, K, V): d_k = K.size(-1) scores = Q @ K.transpose(-2, -1) / (d_k ** 0.5) weights = F.softmax(scores, dim=-1) return weights @ V
```

## Transformer: Multi-Head Attention

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## Positional Encoding

Positional Encoding is a technique used in Transformer models to provide information about the relative positions of the input tokens.

Transformer models use positional encodings to process sequential data.

There are two main types of positional encodings: learned and sinusoidal.

Learned positional encodings are used in models like GPT and BERT.

Sinusoidal positional encodings are used in models like GPT and BERT. They are learned during training.

LLaMA, Qwen, and ChatGLM use learned positional encodings.

## Feed-Forward Network

Transformer models use a Feed-Forward Network (FFN) to process the input tokens.

The FFN is defined as:  $FFN(x) = \text{ReLU}(xW_1 + b_1)W_2 + b_2$ .

The FFN is applied to each token in the input sequence.

The FFN is part of the Transformer Block. The block consists of a Layer Normalization layer, followed by an Attention layer, an FFN, and a residual connection.